

ActInf Livestream #023.2 ~ "Embodied skillful performance: where the act

Livestream | 1:39:19

greetings everyone

welcome to actin flab live stream number

23.2

today is june 15th 2021

and we're going to be having our dot 2

discussion

on embodied skillful performance where

the action is

and we'll see who else joins us live as

we

go through the slides in the paper and

also of course

if you're watching live we'd welcome any

questions or comments

welcome to the active inference lab

everyone we are a participatory

online lab that is communicating

learning

and practicing applied active inference

you can find us

at the links that are on this slide this

is recorded greetings dave

you this is a recorded and an archived

live stream so please provide us with

feedback so that we can improve our work

all backgrounds and perspectives are

welcome here

and we'll be following good video

etiquette for live streams

we're again on june 15th having our

second

group discussion on embodied skillful

performance where the action is

so this is sort of our jumping off point

and let's see if we can

make it a skillful one today the goals
are really just to be
discussing and learning to be hearing
the questions and points that people
found really interesting when they were
reading the paper as well as
see where our discussion can
spontaneously
go as of course participants join
and leave and hopefully we get some fun
questions from the chat
and we're just going to walk through the
paper
and the slides go where we need to go
to hopefully connect some dots
so with that being said let's just go to
the introductions
and warm-ups and we can each just give
an
introductory thought on the paper if we
would like
or something we liked or remembered
something we're looking to
resolve reduce our uncertainty on by the
end of the discussion
so i'm daniel i'm a postdoctoral
researcher
in california and all passed to dean
hi i'm dean i'm in calgary and
i'm becoming a more
embedded personality in this active
inference lab
world and i'll pass it over to stephen
hello i'm stephen i'm based in toronto
i'm doing a practice based phd through
the uk i'm interested in um
participatory theater community
development and ways to understand

how people interact with each other uh
in groups and individually so
this paper is quite relevant as it opens
up some new avenues in that
and i will pass it over to
is it is dave here dave hello dave
yeah dave if you can speak otherwise
no worries maybe you're just listening
in
so maybe just a quick question stephen
where do we
see the connection between skillful
performance which is often thought of
like motor and bodily
how do we connect that to social
practices which are
amongst bodies and use verbal behavior
or
symbolic behavior yeah i mean
this is this is kind of a question i
think that's arising more and more as
people are trying to question
psychology or community psychology which
works on the individual
or sociology which works on this sort of
mass of people
but when you want to work with
intersubjectivity and sort of small
groups
and understanding the dynamics that are
going on there
you could kind of ask phenomenological
questions about what it's like
to be in that state which is kind of a
bit more
direct than say someone's psychological
perspective on it but it's still not
quite working so then you say well let's

get into action here let's get into
what's actually happening in and amongst
people
um and often if you then jump into
action that's another abstracted
area but if you're in the area of
practice
um you're kind of in a in between
you know it's something that people are
doing something that people are doing
between each other
and but you're still able to talk about
how it's happening
how it feels um so i think that um
that this this question of practice then
you know takes you into the sort of
areas that get used a lot in arts
arts practice theater practice all these
types of areas so
um i think it's quite a a nice nexus
that's normally a bit impenetrable cool
it makes me wonder do groups
have skill or is skill something that
only
bodies have so you see some football
team or basketball team do some very
coordinated play
or two dancers that are improvising is
that
able to be reduced entirely to the
skilled embodied performance
of individual bodies or is there
a way in which we could talk about a
group embodying
skillful performance especially because
so many performances i mean except for
what stand-up comedy a few other things
their group and then even those

individual performances are in feedback
sometimes
with a group it's just different roles
um that's a good point actually i mean i
think that's really the exciting
field of practice that's going to be
growing over the next while
and i i sense that it's almost
impossible to get that from a snapshot
you have to have something that's been
progressing over time to get
or let's just say i don't know whether
that's true but i
i sense that there's more use in
understanding how things change over
time
and getting the dynamics than there is
trying to analyze a snapshot
uh dave if you have any comments you can
unmute and i'll
see it or dean any sort of just opening
thoughts how does this dot 2 meet you
today
well i'm going to come from it from a
little different angle i
so i think that word between is a huge
is it it's the thing that that befuddles
us and yet it's the thing that we
probably want to know the most about
and um in this idea of
embodied skill for skillful performance
i was going to actually ask the author
some questions around how she relates it
back to sort of
the poor the parts of the brain that um
resolve these problems so is it a
spatial relationship thing like what the
hippocampus

tends to provide us some insight on or
is it a
caudate nucleus thing which says that we
sequence everything out and we turn left
we turn right we
release the ball and i was just curious
about which
because those two those two things work
but when one is on the other one is off
there's kind of a binary relationship
between those two parts of the brain and
so i wanted to kind of figure out
how that that toggling back and forth is
part of
this skill that we are witnessing
nice point there's um the more
cognitively driven versus almost the
more bodily
driven and then that handoff is where we
see
skillful performance and that's where we
see choking it's where we see all these
dynamics between cognitive processes and
action processes
okay we see another dave dave if you can
if you want to give any thoughts just
feel free
um yeah oh just listening and thanks
nice
right on that's the real that's the real
og way to listen in
in the jitsi rather than in the live
stream
um yeah so we'll see who else
joins in today but that's a great
question about the brain pieces and then
i always think about other species like
if insects can have

embodied action and that seems pretty
fair to say
can they have skillful action and how
would we know
and then if they do the layout of their
brain and body
is so different that it wouldn't map on
you know they don't have a hippocampus
they don't have a prefrontal cortex they
have other
brain regions so we don't need to get
too
locked into one implementation of these
processes based upon human neuroanatomy
when we know that there's other neural
anatomies
that can deploy similar
processes so it helps us
keep an open mind the multiple
realizations
of active inference or of embodied
skillful performance because of just the
diversity of evolutionary systems
let's take an overview at the roadmap
this paper has a form that we've
seen several times before which is
in the first half taking
some non-active inference body of
research so in this case it's skillful
performance
and optimal motor control theory and
investigating with a critical lens this
previous body of research
to understand what is it really saying
or what are the assumptions
so in this case there's this
instructionist assumption
implicitly or explicitly that's drawn

out
through the exploration of previous
approaches to skillful performance
and specifically the motor
representations and motor commands that
are implicit
in optimal motor control theory so
that's
part one part two
goes from optimal control to predictive
coding and
active inference and last time we talked
a little bit about similarities and
differences
as you go from optimal motor control to
predictive processing
to active inference what is generalized
as you move from a to b to c
what is different what kinds of issues
might you stick or
introduce and then in the
last sections of the paper motor control
is positively reframed as this
interactive
sensory motor engagement with the world
which puts us very squarely
in the domain of active inference rather
than in the instructionist domain of
optimal motor control theory
greetings blue so that's kind of a
general pattern that we've seen before
where the first third of the paper is a
background on previous approaches
in a critical lens the second approach
shows some sort of a transition
from theory a to b to c and then by the
third part
there's a positive positivist argument

as to why active inference is
more in line with how things are or
might be more useful
uh steven go ahead
so yeah further to our conversation
earlier um
i think this actually this this showing
the instructionist and then the
interactive
approach is really useful to sort of
have a plausible way for
affect and social dynamics
to interplay with action because when
it's under an instructionist model
it is kind of hard to imagine how that
kind of
mechanical almost message passing type
of approach can
integrate into our sort of social
dynamics whereas
um the interaction in this approach
does give that so i think that's quite
good
it can frighten people off because it
brings in this other
philosophical piece which i think a lot
of people would like
running away from at first but actually
it does
offer a more humorous humanizing or
biological way to engage
cool let's just go there blue of course
raise your hand anytime you want to add
something but this
instructionism versus interactionism
i think is a big piece that we'll walk
away from this
uh 23 width it's something that we can

take on in the case of embodied skillful performance but i think we're going to be able to apply it to education we're going to be able to apply it to team collaboration and so we can definitely return to this distinction between instructionism and interactionism so steven just from you know your experience with drama or other kinds of work do you think people are how do people implicitly use instructionism versus interactionism or what kinds of scenarios uh support or constrain instructionism versus allow for interactions to emerge well i think what tends to happen is in the models that people tend to use they're kind of either in one camp or another it's like so mostly it's almost implicit that there's some sort of interactive stroke flow it probably goes even further it's sort of some sort of flow in the sort of performance arts world they talk about flow and ease and that type of thing and what's also interesting is i know they talk about this in the michael chekhov acting approach is there's always this battle between ease and form so a dancer gets more form they get more technique they get more proficient their technique but they might lose their ease that you have as a beginner

so you're forever trying to add more
technique
without becoming this kind of rigid
autumn automaton
you know so i think there's there's that
kind of side to it and
and that that whole area they just go
off
and it's it's kind of it just takes care
of itself but it becomes a little bit
um of a silo in a way
and then you've got people working in
more straight activity theory
veikovsky kind of much more like
what activity happened when um and that
they they they are also in a bit of a
silo
so i think that you tend to find that
and
different different areas they they are
a little bit siloed in their approaches
um to sort of get around some of these
um
challenges um so i think that
the way of trying to break
beyond what you can analyze which often
you have to look at
instructionally because you kind of have
to break down things into steps
but that doesn't mean you need to keep
it in
a stepwise form in terms of how we
understand
what people really do so there's this
kind of i think there's this kind of gap
that this will help to make plausible
interesting there how we capture
the utility and the apparent advantages

of instructionism

some of them are written here like the

ability to separate different

instructions that seems

pretty important if you say well how am

i going to cook this multi-stage dish

and someone just says you know here's

all the ingredients and here's all the

behaviors but i'm not going to tell you

what order to do what in

if we can't separate and sequence events

then it's going to be hard to carry out

a performance correctly you can't just

switch the order of musical notes and

the sheet music

you can't just do things in a different

order so how do we get

some of the features that we're looking

for

like separability and modularity and

take it into a fully interactionist

framework one distinction there

that the authors raise is like strong

versus weak instructionism

and they'll um end up continuing

that both of these forms have similar

underpinnings

but there's sort of uh a continuum

between thinking that the instructionism

is completely its own neural

representationism

it's kind of like a mott and bailey type

argument this is the strong

instructionism

thinking that it's totally just the

neural representations

that are entirely specifying on their

own

the movements that have to be performed
versus
a more dynamical coupling version of
instructionisms
so that'd be like you know when you're
tapping the gas it's like an instruction
to the engine but it's not the entire
um go faster cue
given the background of ecologically
normal processes that enable them to
play
this role so that's kind of interesting
um blue any sort of overarching thoughts
or anywhere you think it'd be fun to
jump in
no i'm just trying to figure out where
i'm at still
indeed um well there'll be a few other
places like a few other notes that we
wanted to discuss dean
mentioned um one thing maybe we can even
have a dot three
and find out what a dot three would uh
mean or dot two point
dot two one or something like that dot
twenty one um so one piece was about the
the neural elements because in the
figures
we do see brains
however um these are quite schematic
it's not saying that these are the brain
regions where
these types of things happen which can
be a little
uh ambiguous because it's not actually
an anatomical
representation yet it's superimposed on
top of a brain

um okay so one piece we wrote down was
about the neural
and i think there was a few um yeah oh
yeah stephen go for it
yeah just tying this as you mentioned
about the streak the weak and strong
um it's kind of interesting because
that's kind of trying to give this
scientific
feel but it doesn't feel very
satisfactory to
try and have to say it in terms of weak
and strong because you're kind of in
this
well what does that really mean and how
does it shift between the two i mean i
can understand what has been said
but um it's it gets into this kind of
fuzzy version of instructionism so i
i think that kind of sets up some of
these questions that as soon as you move
away from
a strong instructionism um
it becomes harder to hold it together
using the word instructionism
so i think that's it's quite a good
illustration of that
because it's sort of unraveled you can
see how the second paragraph's a lot
longer than the first one basically
yep great point and it's also a common
approach to basically rigorous
philosophical argument instead of making
an argument against a total category
you introduce a distinction in the
category and then
you might find that either one of those
categories has some feature that the

other one doesn't
you know instead of being against all
wars well there's the good ones and
there's the bad ones and then here's my
distinction for just the bad ones or it
could be that by making that distinction
you then can say actually this holds
true for both
of these categories or subcategories
and so that's how i've made my claim
against all x
is by first breaking it down and then
they do again
find that there of course are
differences between these two
forms of instructionism but that they
end up taking on some of the same
assumptions
um cool there's probably a few other
pieces we can explore
this is the section in the paper that
i'm
sharing on the screen that says the
formulation
of sensory motor control in terms of
optimal motor control theory
heavily hinges on two different but
highly interconnected assumptions
so let's see what the assumptions are
because it's probable that
active inference is going to be
contrasting or doing something
different so one is the central
specification of descending motor
commands
and their efferent copies in the form of
detailed low-level instructions for
control

of the motor plant specified in terms of
an intrinsic
frame of reference i.e extension and
contraction of muscle
fibers so that's that instructionism
and then two a separation of forward and
inverse models
operating on complementary aspects of
action planning and executionism
so the first piece is that
instructionism that what's being sent
descending or towards the body are
instructions and the second piece is a
separation of the forward and the
inverse
models operating on complementary
aspects of action planning and execution
so the forward model is like sort of a
planning
model and then the execution is like
okay given what i planned
how am i going to execute that and those
are two different streams
in optimal motor control and so we've
been talking about how
interactionism we can explore how active
inference
deploys interactionism but in
interactionism we have a contrast with
the instructionist point
one and then through a unified
generative model in active and the
reduction of uncertainty in a unified
generative model
we're going to move beyond this
separation of these two
forward and inverse models stephen and
then

anyone else or any questions in the live chat

yeah this is quite helpful it gives that realization that optimal motor control theory it could

it can be very precise you could imagine someone making a robot because you can measure everything you've got access to all the data

and you could make something within like a thousandth of a

millimeter precision which humans or animals can't do

but it it does have a bit of a

limitation in terms of um

being able to integrate and i think that

i don't know if you've seen any of the papers around

the idea of fields being around the robot's

joints i have like a field around the

hand a field around the

and just move into the field rather than trying to

work out what the actual must like the

actual joints should do

and they they out competed other robots within like

two or three months so the same idea of working with

some sort of more fuzzy um

dynamical field

actually and just working out what works most

efficiently or best even though you're

probably adding a lot more fuzziness into the actual

understanding of the measurements

actually has

a lot of them has a lot of benefits in terms of

learning how to do something quickly

okay i think there'll be a few more cool

points to return to on this notion of field

but i think it's also helpful here to

look at how they distinguish

the forward and the inverse

models so here a forward model

takes a system from an intrinsic to an extrinsic frame

predicting the effects of different movements

using musculoskeletal plans specified by

neural activity so again that's that

instructionism that's because we're in the optimal motor control

so forward model is basically plans that can be cognitive into

implementations

that's the translation into motor

commands and

their consequences on the other hand an

inverse

model builds motor commands by inverting the causal change

so first that inverse model leverages a value function of states

this is where we get the inoptimal motor control

the non-negotiable introduction of value as a central

heuristic states have to be assigned

value because value is what's used to determine

which plans to select and that

leveraged on the value function is used to form a mapping from desired target states which again is built into active inference in a sense through the preference over certain types of states but the desired target state has to be specified in a slightly different way as an actual preferred state rather than over a preferences over states in this optimal motor control framework and that's in the coordinate system based upon the external consequences of movements so i want my my arm to be at a 90 degree angle and then that is translated or it's mapped to a set of intrinsic coordinates in the space of muscle fiber activations like my arm is at a greater than 90 degree angle so that needs to map to activation of bicep firing patterns so that the arm can contract so that the angle is smaller and that's why frameworks based upon optimal motor control are sometimes characterized in terms of force control because all the motor commands specify is muscle forces and joint torques so that's kind of the macro to micro mapping is we have preferred states and the question is there's so many degrees of freedom with the joints so how do we go from potentially a broad set of preferred states or

acceptable states
into a specific pattern
of muscular activations so how do you go
from
make your hand just rest where it's
comfortable which is a relatively broad
sort of plateau of accessible states
into
a position that each joint is going to
be at equilibrium
with and the two of these
assumptions that we've been exploring
here so again
one the instructionism and two the
separation of forward and inverse model
that architecture induces the assumption
or rests on the assumption that value
valuable states is what causes action
and that's what that
first in 2011 paper introduced
is sort of flipped that on its head
which is that
well um in models from omct
sequences of action are selected
according to the value function of their
states
so value is something that's cognitive
and then induces action
but that gets sort of flipped
in the 2011 perspective because value is
a function of the states
not a cause of them so
as written here value is an attribute of
states that are caused by movement it's
a consequence
not a cause and when we get
loops you know which one's a cause of
which

something that's locally a causes b but
then
you pull back a level and maybe the
relationship
can also be seen as sort of the other
direction or
maybe with feedback systems or
embodied systems we move beyond just any
single linear framing of what causes
what
but regardless of the direction or
whether value is an attribute of states
or it's a consequence or it's a cause
the point is
there still is this central value
parameter in optimal motor control
theory
and that value parameter which is
familiar to reinforcement learning like
learning preferable states based upon
their predicted value
that's something that we're going to be
able to build on
in active inference by pursuing not just
value
but also allowing for reduction of
uncertainty
in a generative model which can include
pragmatic or value-driven elements but
also elements
that include curiosity like drives
steven
and then anyone else and this ties in
with kasper hess
work with affect you could imagine that
the affective states are kind of
generated
on the fly at different scales so as you

start to move maybe slowly as you
practice
you see that these value states are kind
of almost
revealed and not necessarily available
and then maybe there's a state
for once the whole action is complete
and that gives you
you know enter that value state which
maybe only appears
as that um degree of surprisal
is realized and it's always going to be
slightly different
each time you do something so this this
is quite cool
i've moved here to that
kind of vector field representation of
the difference between
value only learning and
performance here's our robot who's only
interested
in kind of taking the ruler putting it
on the hill and then going in the
direction of the ruler going uphill
that's the sort of scalar approach to
pursuing peaks
in contrast we have that
peak climbing or hill climbing element
in active inference as that irrotational
or longitudinal current
but we also get this solenoidal element
so i'm wondering
talking to people about performance for
example
or training in some way how would it be
different
to promote this sort of value and
equivalent

flow model rather than a
comparative reward driven model like
if a learner is constantly assessing
whether it's a multiple choice test or
whether it's baseball pitching or
something like that
if a learner is always assessing the
value of states
and everything is framed as value
maximization and you're going to want to
be doing
accurate states you want you know the
pitch you want to be getting your
pitches more accurate because more
accurate is more rewarding
i'm just wondering how it changes
training or the experience of practice
to know that there's times when one is
pursuing value or that
in the long range pursuit of value can
be achieved through
this rotational or exploratory element
blue and then anyone else
so just talking about pitching
specifically like yeah
you can you know hit the bullseye right
if you're
gonna pitch you know pitch a ball you
can hit the bullseye you can hit the
target get it right over home plate
but like you never explore like what
about like
spinning the ball what effect will like
different spins that you put on the ball
have
on you know the way that the batter hits
it so like the accuracy versus like
the finesse of it and i think about that

in terms of dance also like if you're
not
if you don't explore like you know what
it feels like like if you're only trying
to accurately make a movement
you're never exploring like the finesse
element of it so there's no like
room for perfection or like ultimate
mastery i think
and where does skill become mastery
it's sort of like at least in chess it's
the the masters who
appear to challenge some of the basic
rules like they don't castle early
because there's some deeper plan or a
jazz musician
who moves beyond the framework of
regular
timing to introduce some other element
so that play element becomes especially
important
it's almost like if you never learn how
to do it
correctly or normatively you can't throw
the baseball so you got to get it over
the plate
but then the mastery level after
performance has been
demonstrated in a sense involves
inventing a new way to pitch like
there's you know a
fraction of baseball pitchers who throw
sidearm or throw knuckleball
all these different sort of
unconventional approaches
that aren't the value function that they
learned initially
or maybe it was their biomechanics and

that was the value function that worked
for them
and some other value function just
wouldn't have worked with their
body and brain steven and then anyone
else
yeah thinking from the perspective of
coaching because um
so my supervisor works a lot with
systemic coaching and performance
coaching
is this this becomes a more unifying
approach
it might be for specific things
optimal control theory is quite useful
for just
unpacking something but if you want to
look at performance coaching
more broadly or developmental coaching
more broadly and how they relate this
could be useful and i'm just thinking
about
christian ericsson the you know the
soccer player i know if you heard he
collapsed during
the euros and uh his heart stopped and
all the team were around him trying to
and then they had to play the game
in about an hour's time now
their ability to perform
um was going to be very hard it's not a
performance coaching
question at that point there's these
other dynamics going on
and i think that this this ability for
active inference to
at least unify how regimes can change
and how it could be the environment

the system the the emotions all these
things are quite um
quite useful
thanks dean yeah
yeah i appreciate what you're saying
there steve and i i am
aside from the from the step-by-step
part that i think the caudate nucleus
tends to to perform when we're when
we're trying to
climb that little mountain there
robotically
it's another factor i think that's kind
of interesting and that's um
if it's embodied and it's skilled
there's probably
three kinds of math at play there's
materialization the how
sort of the engineering of it there's
the what's in what's out
what's supfliferous and what what needs
to be
an element within the performance and
then i think what this paper does
in the interactions piece pieces it
brings the the probably
into the conversation
and i think you need all three kinds of
calculations in order to be able to get
a real grip on
on what's going on maybe not an optimal
grip at least
a fuzzy grip on what to do next i think
if you
only have two of those elements you've
got the perfect conditions for an
instrumentalist approach when you add
the probably

whether it's erickson um
what did he text or email all of his his
teammates and said
go finish the game we gotta we gotta
finish this game
and i think steven's right you can you
can give an instruction to your
teammates
and say i'm fine but you just had a
cardiac arrest
and we probably aren't going to be able
to wipe that from our memory
in that moment and so it's a little bit
more complicated when we add probably
but i think we need probably in order
for it to be interactionist
so that team like in the middle of the
game
the personnel of the team changed
in the individual story of motor control
it's like but that's not between you
and the soccer ball so why should it
change performance but when we
have the deep generative model of the
individual that includes affect that's
one way to get to
the loss of a teammate influencing
performance and the second way would be
to actually
genuinely allow teams to be skillful
performers
in the sense that there's a distributed
cognitive process
on the team just like there is in the
body and so
they actually changed and whoever steps
into
that person's position on the field

they're gonna have a different approach
so it'd be like switching out you know
one arm for another arm
or one forager nest mate for another
nest estimate
it's going to change the emergent
properties of the group in a way that's
an explanation at the group level rather
than
requiring to come back to just the
optimal motor control
value driven instruction guided of a
body
blue
yeah just the probably is like what
gives active inference its power
right like probably if i don't eat
breakfast i'm going to collapse in the
middle of a of a field you know or
playing a a game or whatever um so so
when you have that
that probably it it allows you to guide
all of your
actions toward a given intention i think
thanks blue steven yeah and i also think
when probably
because probably often it's still in
that venue
my world you know probably i will get
this game or that game
when the word uncertainty is put in
uncertainty is a much scarier word than
probability right because it's like
it has that risk component um so it's
really interesting that you say
that that that um we tend to say
probably everything will be okay but
active influence is like

you don't know it's all it's all unknown
to some extent here's
here's another nice piece from the paper
that
i think can continue this discussion
about probability and
maybe also how active inference reframes
it so
the physics of flow embedded in active
inference now
it's an ongoing discussion exactly the
formalisms that get us to
a physics of information flow on sense
cognition and action so other
live streams other days other years but
let's just roll with this
physics of flow it's accounting for
sensory motor behavior to show that
motion in a biologically realistic state
space irreducibly includes two
orthogonal kinds of motion
the irrotational and the solenoidal
component so that's what we're looking
at here
just this is a paragraph that's
representing this piece
instead of just value only we have value
and flow
and again this is key to reiterate
because it's something that isn't
included
in reward scalar representations
so the former that the first one the
curl free
is what allows the flow to climb a
gradient towards more valuable
or probable states now that's one of the
head scratchers in active inference

that preferred states are likely states
under an optimistic deep generative
model
and that is the alignment
that allows us to reduce uncertainty
and by way of reducing uncertainty
achieve
our target outcomes and so when we align
the deep generative model in an
optimistic way
such that we believe that likely states
are preferred ones
that it's likely that tomorrow i'll have
a regular body temperature
it's likely that tomorrow i'll be in
this preferred body position
then we actually get to pursue
value as a consequence of reducing
uncertainty
there's more to say on this but dean go
for it and then stephen
yeah and i think daniel that what that
points out is that there's a dependency
for people who who who want to introduce
statistical
propositions to what's the underlying
situation that we're observing
um there's a dependency on the
derivative math and the accounting math
you you can't you can't figure out what
probably is
without figuring out how and what and
having a
having a a sense of
confidence around what how and what are
so i don't think it's
it isn't just the introduction of active
inference

active inference is also wholly
dependent on a strong
foundation in terms of really being able
to get
from priors through that hidden state
and how do we do that we have to have
derivatives
and we have to have accounts so i think
um
there's a real codependency back to the
between comment
that steven brought up
cool thanks steven
yeah that that that idea of um
things going through cycles i think that
i'm not going to go into ergodicity
again but you know it often comes up is
that's from what i'm understanding is is
pretty much
not present in optimal motor control
theory because it's
it's about you go there and then now you
have a vector
and that vector is a thing and that
thing gets optimized but
you never have that thing i think like
dean was saying there's that kind of
in-between and
you're kind of holding that in between
but but
that only makes sense if you keep
revisiting the in between
like if the you need to have ergodicity
of whatever
i mean that's an open question again for
another day but
the there's some way that you're
revisiting this understanding because

otherwise you've got no way to make
um any sort of information gain
or information extraction unless you
have a revisiting
to compare interesting
um and it's
one of the key pieces again that
we reduce our uncertainty and achieve
preferred states
through that process i'm just imagining
you know something from
organic chemistry it's like you want to
study up cram for your final at ochem
you want to pass
so are you going to take a reinforcement
learning perspective or an active
inference perspective
well the reinforcement learning
perspective is like saying you'll get
the big reward when you pass
so what's the most rewarding problem
that we should be solving or what's the
most rewarding chapter to review
but you don't know because if you knew
then you'd pass ochem
so you're kind of right back at square
zero with value because
you framed learning as a value question
in contrast the active inference learner
would be like oh
sure my preference is to pass how am i
going to reduce my uncertainty about the
critical path
of passing well let's i haven't even
opened the textbook let's go to the
first
table of contents or something like that
like reduction of uncertainty

gives a first step and then it even
allows for
okay maybe that wasn't even the right
first step or maybe that was a good
two-step
path and now i'm ready to go a different
way but i'm trying to reduce my
uncertainty
about how to get to what i prefer
which is passing rather than i have a
priori valued
this future state and now i'm going to
back
propagate value at every single step of
the line
and then i'm going to stress myself out
if this specific chapter
isn't maximally rewarding because
everything again
has to get reduced to value in this
value is everything framework but in the
uncertainty reduction framework
we see sort of this mist of uncertainty
and then regularities and especially
dependable ones
emerge out of that and so we're planning
amidst uncertainty
with preferences in mind trying to
figure out what to do
that's planning as inference amidst
uncertainty
which is just a world apart from
planning to maximize value through time
given
you know what model again of how value
is assigned
it's just a recursive question and value
learning doesn't give us that higher

level answer

for how we should actually go about

planning to obtain values

let alone critiquing value as a

pervasive framework

stephen well actually that's a very

interesting

talk about organic chemistry experiences

at university

that i had an absolute nightmare on my

organic chemistry i

might i've given my textbook to someone

i've got behind on the organic side

and um it was this like two inch thick

textbook on organic chemistry and every

time you go into it's got so many words

it's long names and all of this

and what's going on it's really

interesting because we had this class

with the new

professor and he got there and he got a

few people to go up to the board

and it became clear that absolutely no

one knew their organic chemistry

and no one had seemed to pick up on this

so we were now in the second year no one

knew

they were doing i think it was even the

third year and he said

and he said he recommended a little thin

exercise book

with exercises not on all the different

chemical reactions and all the

structures that just

the key reactions and just to go through

these exercises in this thin book and

basically

that got me through the exam and got me

on track

not and i you know so going for the big
book and the

the memorizing and the optimal control
it was

just work out the five types of reaction
and do the exercises and i think that
speaks to what you were talking about
there you know which ways the electrons
move

and and that suddenly really really
changed everything

there's also some fun drilling happening
of course

all around me welcome back

dave um so this

section right here it's it's a nice
challenge

it increases the difficulty of speaking
but it's fun

this um is a challenge for other
frameworks this idea that the two flows
together

provide a richer framework for
expressing skillfulness

as a process whose characteristic go
beyond a simple gradient of value
accuracy

so we saw that reward is enough paper
recently and some participants in our
lab were kind of curious about writing a
response

because that's the whole argument is
they said hey reward is
enough the kernel can be reward and
that's how you get

all of these intelligent behaviors
provided that you have these second

level
um tweaks on the model or these bolted
on modules
and then the contention of the authors
here
is that value functions and indeed any
motor
scheme based upon functions that return
scalars
are not up to the task of modeling the
variety of skillful acts describing
human behavior
any kind of behavior just human because
by construction
they cannot account for the solenoidal
aspect of flow
so that that's pretty interesting that
when we have to
bottleneck into a single number
we lose a lot of the expressivity that
allows for
explore and exploit behavior to almost
coexist
because we're collapsing onto a number
line
so there's extremely limited
availability for
you know what do you want for your main
dish and your side dish and you just say
seven you can't express two things
with just one number but if you could
say you know seven and three
you could express something that's more
in a two-dimensional space
blue so i think you bring up an
interesting point about any behavior
or just human behavior um and i think
about

you know there's like dolphins that use tools right to crack open like fish and or to crack open not the fish but the um the sea urchins right so there's dolphins that use tools for eating and then there's dolphins that don't and and so is there and like this is like a hereditary like it's it's an example of cultural evolution in dolphins actually so it's you know it's hereditary but only because it's passed on through learning and social learning right so you know you bring up and and then we saw the motor plant right like the motorized plant like does the plant learn where the like how like skillful performance i mean this is a skill for performance on the part of the plant so it's a really interesting question whether you know i mean we think about active inference as applying to all non-equilibrium steady state systems but does this maybe apply only in the human sense of movement movement or what about like tribes of humans like what about like in a nomadic kind of population would this apply there also cool here's another nice piece in this section that i think will be useful for us to learn about so technically it's easier to read when it's already been written when you're

getting the drilling
experience that's when it's the hardest
to skillfully
improvise but reading is a little easier
technically
active inference extends popular
predictive coding models used in
neuroscience
where perception critical only
perception
is cast in terms of prediction error
minimization so that's the classic round
ballard
99 paper that's sort of a mega citation
and there's a huge amount of
subsequent work and predictive
processing or predictive coding
is just looking at the relationship from
sense
to updating of a generative model of
sense so it's a signal processing
framework
and this is something i know that dave
is a big fan of
both he knows drills and hammers i know
that dave is a
fan of talking about the signal
processing
and the audio processing routes of
active inference
so this is just on the inbound just
from sensory information to estimates of
that generative model of that
generative process of the sensory data
the active inference framework
extends this account of predictive
processing to model
motor control and explains action

selection

by appealing to the minimization of

divergence between predicted sensory
data

and actual sensory data so here we have
the exact

same divergence minimization it will
sound better on the live stream

my jitsi colleagues don't worry um it's
the same

divergence that's being minimized which
is the difference between

observed and expected sensory data so
in predictive processing it's about

sense and perceiving sense and so you
want to minimize the divergence between
what sensory data you expect and what
you're actually getting

but active inference adds inference
uh planning as inference into the loop
and explains action selection also in
terms of the exact same divergence goal
of predictive processing

which is again to have your observed
sensory outcomes
come into line with your expected
sensory outcomes

dean yeah i

i one of the things i used to

share with people is so today i'm i'm
wearing my

uh murder of of crow's t-shirt because i
used to study a lot of corvette behavior
and even though i there's
there's still some question as to what
sort of

verbal cues there what instructions

corvids are able to share because they

are social animals and they're
they are problem solvers but you don't
have one corbin standing
to the side as another corvette solves a
problem giving them instructions
so i think it's kind of interesting when
we're when we're looking at this
behavior
question whether we're the only species
that are actively inferring
because i think that there are other
species out there that you can make a
case that
they are doing that probably
form of questioning and answering as
well
um so and and yeah it can be tools
it can be based on on on social
constructs
um it can be based on caching
so deep memory and again there's a
dependency i think
for behavior that uses active inference
to also
some grounding in how and what
so i don't think that active inference
in and of itself
will allow us to to
minimize anything unless we have the
derivative
and the accounts so i think it's i think
it's it's a virtuous triad
here but i think what this paper is
trying to do is get away from the idea
that
all problems can be solved with only two
kinds math
cool thank you stephen

yeah further to that point around you
know
what humans do or animals do in general
is
you could say optimal control approaches
maybe
it's only really humans that can apply
that extra level on top
as this kind of constraining kind of
deductive
idea and every every other organism is
really
working with this kind of flow i mean if
you see a squirrel trying to run up in
a tree and between trees and chase
another squirrel around the trees it's
quite phenomenal
just how agile they are and the speed
they can work at
and um so there's some sort of skillful
performance that they've got
so there's this interesting
thing whereby you've got this we
see optimal control as more
straightforward
but it basically it's almost really
optimal control just
isn't implementable in practical terms
it's almost it's a it's an what you
could say it's a
non-ergodic deductive extra
that humans bring to the table to
explain
the mechanical world that they're
bringing to bear
in some way so it's kind of interesting
it just happens to be that was the first
explanation we had for this type of

stuff

nice so let's think about what active

inference is

and then see who might be doing it or

what might be doing it

so again the core idea is that rather

than

selecting commands the organism in

active inference is inferring what it

must should be doing under this same

imperative of predictive coding which is

to minimize the relationship

minimize the divergence between observed

and expected outcomes through time

crucially this brings perception and

action together in the same functional

profile

we have them on a current sort of

trading uh

market almost like we can reduce our

uncertainty through

updating our model and we reduce our

uncertainty through action

so updating an action or action and

inference

are put on common footing and it also

explains

some so this is the sort of

computationalist or

um algorithmic level is we're making a

sensory motor model

that doesn't just have arrows connecting

pieces ad hoc and say look they're

connected because i drew a picture

we're bringing perception and action

together

in a common modeling

framework and also on the neurological

it explains some of the similarities
between the functions of sensory and
motor cortices
so that speaks to dean's points earlier
and to a lot of these brain regions that
might have been discovered
or initially characterized as having a
more cognitive or a more motor role but
then it's shown that they play
other roles and so you know cue the
endless clickbait oh this brain region
associated with this
also does that should that be surprising
should that be salient or
should we have another view where that
is to be expected
now while this move from a problem of
control
so optimal motor control that's the
action side
and then there's so pure action side
optimal motor control
tends to not concern itself too much
with uncertainty of measurement or deep
generative models
and then you have the pure sensory
processing side that's kind of
predictive processing
this move from a problem of control to
one an inference in terms of active
inference does not make the problem
mathematically
easier in and of itself but it offers a
different model of skilled action which
respects the neurophysiological evidence
so in other words we're taking the sort
of problem of going from
sense data to a generative model the

predictive processing model
and then the forward model from
internal states to action selection
and it's easier to of course
subset to just one of those directions
and say that one has a tractable model
but active inference is combining both
directions
or integrating across these two
different
ways of like looking at sense in action
and saying let's have
one model of sense in action with a
common
functional profile as they describe it
here now
which systems have it slash do it
that's the realism question and the
instrumentalism question is
which systems can we model as if they're
doing this
so i think that the instrumentalism
question which as always it's about
us so it's easier to answer is the kinds
of systems
that have sensory motor input output
dynamics so everything from a
self-driving car to
different kinds of animals we could
model as if
they were getting input generative model
output so the things that people model
with ooda
or the that classical sandwich of
cognition
those are the kinds of systems that are
apt to be modeled with active inference
the kinds of things that people were

modeling with optimal control
going from the model to action and the
kinds of systems that people
were modeling with predictive processing
from
sensory inputs to generative model
both of those and ones that especially
are doing both those are some of the
most
amenable uh systems for active imprints
stephen
you know i was wondering this this idea
of functional profile
is i haven't heard that one before it
might be another one to think on the
ontology
that's been talked about how have you
seen that being used anywhere else um in
terms of active inference
functional profile
good question it doesn't seem like it's
used anywhere else
in this paper and i
don't off hand recall
what other papers it might have been
used in but that's a great point
something to ask about
um i think that the notion of just
function
is a key one to follow and
uh what what
how how do we follow a function yeah go
ahead stephen
yeah i suppose because this is talking
about
actions skillful performance function
starts to it's thinking about functional
as in how to achieve an

action of skillful performance so maybe
that's how it comes in here
whereas a lot of some of the other
papers there's no moving parts so to
speak it's all information based
in in terms of um but i think it's
it's interesting in terms of um
what that does or what it's trying to do
it's another way of trying to speak to
something which is a bit bigger you know
that's one thing i've noticed inference
normally things are trying to define
what everything is this one it gives
these it
gives these pullbacks they're like okay
we've got a functional
profile now that allows us to
um do xyz but like i say it does provide
that challenge of
trying to fit that into these other
pullbacks
that we seem to be keep finding
appearing
like you know the effective charge
that came up in one and so yeah i think
that's a kind of an interesting
question about how that fits in
with other types of models or
ways of translating between spaces or
frameworks
dean thanks yeah so i wonder
so let's just take that to its its next
logical place so
let's start with a functional profile
not necessarily work back but
is there some way for us to be able to
identify
what's in that profile that would allow

for active inference and should we
is that something that that is that
dangerous or is that
legitimate
nice question
here's a little bit where they're
getting at this realism and
instrumentalism
discussion so even some of the most
forthright defenders of the active
inference
framework assemble have backed away from
making such strongly realist claims
about the causally
efficacious character of generative
model so here is where we see
um hardcore realism so it's nice you
know and
as we work towards the terms and the
ontology how can we classify
different senses or uses of terms
first in 2013 has advanced the view that
an agent does not have a model of its
world it is a model
now that's an interesting claim
um but the authors bring this up
by also qualifying Friston's use of
model from a different paper
and saying crucially if we accept the
generative models of the active
inference
frameworks not sure if this
is the if there's saying you know
usually it's framework but
are will there be multiple active
inference frameworks in line with these
proposals we must embrace the idea that
generative models are causal in the

sense of an agent's inactive attunement
with the world
rather than part of the causal
efficacious machinery used by an agent
so that's an interesting claim blue
so i i was thinking about like you know
that an agent doesn't have a model of
the world but it is a model and like
i think about that in terms of people
that are that get
have traumatic brain injury and then
have like
dramatic amnesia like can't remember how
to walk or
talk or eat and have to like relearn all
of these
things right so so i think about like if
we don't
i mean in that case it seems like the
model
like you have a model and then you
suddenly don't have a model
but it's like if you are a model
are you suddenly not a model or is your
is that like a temporal
disruption i don't know like so that
that to me i mean
and this is an instrumentalist realist
debate i realize
but where do you you know if you are a
model is it like a temporal disruption
and suddenly you have to start over
or what does that that look like in in
those circumstances
nice question stephen
yeah i mean in a way i suppose when
someone has to re-learn
that the fact that they are at some

level their body is part of being that
model that gives them
access to either some priors
or that they call it the morphological
computing capacity the body's capacity
to then
form those priors because um
you know if if however certain parts of
the body are
like with a stroke that can be kind of
challenging because that
maybe the actual way the body moves
is compromised therefore the ability to
use it to recreate
is is diminished i mean the question of
how much it's diminished then becomes
i don't know if anyone could quantify
that but it's interesting
it's a good point an instrumentalist
would say well yeah the brain
isn't you know per se doing active
inference we're going to model it as
such
but then when there is an injury to the
physical system
then we say oh we're still just modeling
it with active inference we're just
changing our active inference model or
has the actual
system itself changed so will we just
always say that even when the system
changes
in the physical world we're just going
to update our
model and just always stay in the
instrumentalism land or
are there times where changes in the
world

make the instrumentalism verge into
realism as if it's actually describing
something about the world that's going
to be a very delicate
walk and they they
are arguing that again if we think about
these generative models
as purely formal and epistemic tools
so just like a linear regression you
know you can do a linear regression
on gdp or on some other feature you can
do a t-test in a double-blind study it
doesn't make the people a t-test
it doesn't make the gdp a linear model
so it doesn't make something active
inference
just to use it in this statistical
framework
that's what's meant by the formal and
epistemic tools
then this framework is we argue
compatible with a sensory motor approach
to understanding skilled action
it's interesting to me because the
sensory motor approach to understanding
skilled action
itself could have a realist or an interp
or an instrumentalist approach
the instrumentalist sensory motor would
say i'm not even saying this system has
sense in action
i'm just saying we could model it as if
it had sense in action
whereas the realism of sensory motor
would say this system
actually does have sensory and active
components
and also this is another important note

the sensory motor view
reflects a spectrum of ideas which
includes simple open loop
sensory motor correlations closed loop
systems regularities given a goal and
optimal sets of regularities according
to a certain performance
metrics these can be understood using
the tools of dynamical models of
cognition
capturing the brain body environment
interactions in terms of dynamical
systems
as opposed to assuming the agent is
involved in symbolic computation
so it's almost like we see a lot of
separation of
uh the isolation of agency or the
delineation of agents
with agent in the niche and then the
niche itself so we see like agent
and external internal states and
external states and now when we think
about brain and body
that's one sort of inseparable dyad
and then it is also inseparable from the
niche
so how are we going to take all these
dynamical systems which have different
physical and causal connections amongst
them
brain body and niche and
work towards a framework that's actually
going to
say what we want to say and be useful
for what we want to do with these kinds
of models
like reducing uncertainty about how

organisms will
act or designing environments that work
for
different sorts of agents how are we
going to do that effectively
and drawing from what we know about
dynamical systems more broadly
rather than bring on assumptions such as
that the agent is performing
symbolic computation which comes along
with the instructionism and the
representationism and all these other
pieces

stephen you know this is a useful
uh sentence here to show that dynamic
that's happening between the kind of
participatory sense making world um
and the sort of inactivists and
the inactive inference

world um because there's there's these
useful distinctions that have been
coming up

in an activism and this participatory
sense making

yet they struggle to give the ways to
actually

calculate how that comes about outside
of these kind of metrics

that are looking from the outside of the
system so

um this this is kind of

quite a dance that's going on but it
seems to be yielding some

good um progress for active influence
because

they're laying some of the ground for
active inference to then see what could
be plausibly

uh modeled so that's quite useful
yeah another point to that and also what
you said
earlier um
they're suggesting that active inference
is going to extend
costly linear accounts of motor control
which tend to focus on the contingency
between
new actions and their consequences so
the loop
there which is hardly a loop it's
actually a line goes from
action selection to a new
state like i want you know i want to
grasp the coffee cup
how am i going to grasp i grasped that
is
the domain of many previous models which
is like from some sort of delta
that needs to be accomplished to
achieving
that action and that's where the model
ends
active inference in fact proposes a
complementary view
where predictions of expected
proprioceptive states
are not just seen as passive reactions
to new motor signals
like in the coffee cup grab example but
as
also triggering adjustable dynamic
reflex arcs to generate new actions
so here instead of starting from action
to a new proprioceptive state like i
want to be in a different state
so um i want to select an action okay

now my elbow is in the correct
predicted proprioceptive state so we're
good to go
here we're starting with the updating
of our proprioceptive state moving to
action selection
inferring our state so action and
inference
like the two-stroke engine are now
working together
in a way where we can pull out a little
local stretch
that has the same functional features
as this linear causative model above
because there's going to be some stretch
where
that action was performed and where the
functional characteristics were carried
out
but we're embedding that in a
deep temporal model and the temporal
depth of this model confers a more
active anticipatory role to
proprioception
now seen in a causally circular model in
line with the inactive and embodied
approach
so that's that sort of rummaging around
that gives
an anticipatory and active role for
proprioception
not just a state estimate role for
proprioception
so that it can get relayed to the
central headquarters
and sent back as an instruction it's not
like a central intelligence model
of action selection it's like an

anticipatory

deep temporal model that allows sense

to be doing something slightly different

stephen

um just a question when it says and so

it says new action so new action in

active influence is

the new well guess

i suppose of what the action state is

and and then the new proprioceptive

state

this is interesting thing with

proprioception

is is that almost a guess as well

like based on a mix of um

sort of intrceptive data

and other data

or data but other ways so is it like a

different types of guessing

or is one of them action ones

still being seen as perception in that

loop

good question i'm sure there's a lot of

ways to go about it

but one thought is even perception

is inference that's predictive coding so

our

visual perception is clearly a generated

inference there's no blind spot there's

resolution and color in the periphery

where we know the anatomy of the eye

doesn't directly support that

and so on there's smoothness even though

our eyes are constantly secating

so whether it's a proprioceptor

or whether it's visual input that is

also

inferred as part of the deep generative

model

and so when there is ambiguous visual

input

or ambiguous proprioceptive input for

example

that inactive inference is an

opportunity for uncertainty reduction

rather than getting out over our skis

within saying whether that's a valuable

or

a less valuable state we actually

don't need to judge the value in a

scalar framework

of ambiguous versus unambiguous states

because we can just pursue uncertainty

reduction

and then also cast planning as inference

and perception as inference does that

make sense stephen and then blue yeah

that's that's really useful and

i think it's like what you're saying

there that also brings in the

the idea of temple scale so there's with

vision and with this

there's what's going on below

at greater speeds and flux than we can

know like the circades and the way those

circades then get put together to be

this kind of joined up smooth perception

of vision and

this this is this new action new

proprioceptive states that's almost kind

of

at the speed of everyday awareness

you know and once things become longer

and happen over the period of like

10 minutes intervals you can no longer

track that in your working memory or

your working phenomenological
consciousness
so this is kind of like the goldilocks
zone in a way where we're able to
when it's talking about new action new
proprioceptive states it's kind of
sitting at a kind of um an
averaged out perceptual
awareness in a way that we can engage
interesting point blue
yeah daniel you already like kind of
said what i was going to say right after
stephen but
um proprio reception really is is the
expectation
like where do i expect my body to be
where
like instead of asking like where is my
leg it's like where
is my leg supposed to be or where do i
think my leg actually is
and you're not always right right like
in the phantom limb
um case and even like you know if you're
stuck with a pin
right like so and this is not
proprioception this is just
generally like where your nerves are so
some in some places they're very close
together
and you can infer with great accuracy
where exactly the pin was poked into you
but in some other places in your body
the nerves are more
far apart and so you can't tell even if
the nerve even if two pins you put a pin
here
you poke a pin here you touch the same

place it's like the pin
is here but you're not always right so
so there's there's
this inference that that happens um in
proprioception
as well nice it's like there's
um one mode where you have a very
high confidence prior of where your arm
is located you know it's right by my
side
let's just say and when there's a high
confidence prior
data that don't support that prior are
only going to move it
slightly if someone's very convinced
that something is the case
and they say it's 100 the case that this
is this way well then
you know there's no point in providing
alternative evidence because
they're going to see it in terms of
their prior so in that case of the
phantom limb
it's like there's a high precision prior
that the limb is in a certain location
in fact it's such a strong prior that
sensory data
like proprioceptive data that your arm
is in a different place or visual data
that
for example somebody no longer has that
limb
still doesn't manage to update the
experience
or the generative model of the limb
being there so that's one case
that we can frame in an active inference
way and another case would be like

where's my leg i have low confidence
and now instead of framing this just as
a estimate that there's an uncertainty
on and then we're going to have to tag a
value on

we go from that estimate with
uncertainty to
planning as inference what motor control
or what motor behavior is going to
reduce uncertainty on the leg and
sometimes that's like looking
at it other times it but that's a motor
behavior with the eyes
other times it might be doing a motion
that then like kind of uh gives us some
sensor

data so it's kind of i don't know like
checking your phone and seeing like what
notifications are there when you're
uncertain

maybe sometimes like a shake or
something like that it's like just
reactivating a bunch of sensors just to
pass

say oh yep we're here just checking in
reducing your uncertainty
and so the state estimate even when
there is uncertainty about it
we can reduce that uncertainty through
action selection

rather than just being trapped in a
framework of like

okay i'm not sure where it is and now
i'm not sure whether that's rewarding or
not

and now i'm not sure which path to take
that's going to be most rewarding
we don't go down the reward rabbit hole

because there's a deep generative model
and
for evolved slash created or whatever
organisms those priors are
fit so it's not just like we're dealing
with the space of all possible
generative models these have been
evolutionarily
selected to be functional
in our physical constraints but
underwater or
in a different gravity space it'd
be a different set of priors that would
need to be
uh learned or utilized
in our last piece here what what do you
think were some of the big
pieces that we leave this 23 strange
attractor
with as we move forward like what kinds
of
distinctions or insights here do we want
to see how they apply in different
settings
dean and then anyone else
okay so this is gonna be a strange twist
but that's
that's okay so i came away from the
paper
and i thought about some of the things
that
casper did bring up casper hess did
bring up in terms of
so so what happens when
when we talk to instructionalists like
instructors are all around us there's
people that spend
a great deal of their life preparing to

become an instructor
and we come along and say um yeah that's
great
we know what you do um but we're gonna
kind of
i don't know maybe bring you down a peg
because we're going to introduce this
other
stuff that you weren't asked to sort of
train yourself up on
and i'm not sure if that's a threat or
not
i guess it depends on the personality
who the idea is introduced to but
i'll bring i'll bring up
if you know the movies independence day
and thelma and louise
um i'm going to bring those into kind of
into a
juxtaposition here i think for the
instrumentalists
um they don't necessarily see that there
can be
for the person who's living out the
adversarial
greek myth us against the aliens or
us against the the people who are
chasing us down and trying to get our
our convertible back um there isn't
there is no set of instructions per se
that those people want to follow they're
living the interactionist
state and you could start out with kick
the tires and light the fires all uh
harry connick jr and still end up in a
rapid deceleration moment a la
susan sarandon and and gina davis
and i think for people who are going to

look at this problem
in terms of how do we introduce
interactionism
in a way that doesn't threaten
the top-downedness of how most
most situations where we want people to
be able to take up this
embodied skill skillful performance
without
putting them in a in a situation where
they
behave like aliens or behave like police
chasing somebody down with a helicopter
we're gonna have to sort of
i don't know i don't know if it's a slow
rolling i'm not sure exactly what it is
but those people who are now currently
instructors and instructionalists by
by profession we're going to have to
give them
something so that they don't cling
desperately to
a model that i think the authors of this
paper
demonstrated requires a rethink i don't
think we throw out instructionalism
but i absolutely do think that it's it's
place on the top of the mountain
and now all these little robots climbing
up it
that that has to be reviewed and
revisited and so
i mean as a go forward i don't know how
we put people
in a place where they're comfortable
with that kind of
i don't want to say it's disruptive but
i think it is

but i think we have to be able to come
at this in a way that doesn't make
them sort of circle the wagon
solenoidally and
try to protect something that obviously
is
is up for a bit of evolution before a
revolution
nice points there stephen
yeah i like that idea yeah this
challenge of
having this mountain that we've kind of
ascended
and placed these optimal control
theories on
and maybe it's it's more like there's a
mountain
of active inference and there's a
there's a little rock
a rock structure on the top which is
optical control theory which
we're able to sort of utilize on top of
that you know but it's uh
it's it's a lot more rickety and it's
not the
the main game in town you know the main
game is our intuition
our dynamical um
balancing of all these different free
energy
parameters um but we do and particularly
humans have this ability
to impose this extra
um shack of tools on the top
that we now think is the main game in
town
nice so to stevens then to dean's point
so as to which one's the uh

mountain which one's the mole hill which
one's the ant hill
here fristen says optimal control can be
cast as
active inference with three
simplifications so as we
increasingly clarify our understanding
of these different models
will be able to see which models are
nested within
one another or are alternate framings of
one another
or are you know more overlapping than
different or what are those key
distinctions or what are the variables
where you can expand it to three and all
of a sudden you've gone from model a to
b

so remains to be seen and
by uh him saying that optimal control
is a thrice simplified active inference
along these three specific domains
it does suggest that the bigger picture
might be active inference and then you
get special case
optimal motor control with a value
function as a
derivative model within an active
inference
universe even if these other models were
developed first that's often
the case that a certain model is
introduced and then it's
later generalized in several ways so
that in the future it's actually seen as
a special case model
even though it's actually at one point
it was the only game

in town certainly for a lot of biology
models that's the case
and then dean i really like this idea of
like there's a job title called an
instructor
and if somebody says i'm an instructor
in
you know organic chemistry or in tango
baseball people know what you're talking
about so how could interactionism
be so broadly
participated in that people say yeah i'm
an interactor
i'm a baseball interactor or i'm an
ochem interactor
maybe the job would be called that or
not so how do we
like you were saying not kind of
threaten
or um disrupt their modes but
give a yes and for what they're working
on and
maybe part of it is that the interactor
it's like we're framing their
interactions with their
student let's just say as a cultural
scaffolding
of an interaction that doesn't make it a
symmetric interaction
it's a culturally scaffolded moment
of bayesian model updating and so it's
like
yes you do give instructions that's the
verbal term for what you say when you
say pull
or breathe or okay slow down you can
still give instructions we're not
saying that that is not part of your

toolkit but let's
more broadly look at your interactions
with this dynamical system
as doing what you want to do
and maybe it'd be enough just to push
them to that
edge of the hill and just say
interactions are a broader
way than instructions would you want to
reduce everything you do to just
barking instructions or sometimes you
lead by example other times you
show wordlessly you don't need to frame
your
interactions as instructions recognize
that you're
just dynamically interacting systems and
this a lot of people are working on a
framework that suggests
that this is a broader way of thinking
about training and
might be able to help you on one hand
with the high performance elements
and on the other hand with accessibility
and in making your work more
communicable
not sure about that of course but just
one way to go steven
yeah with the idea of roles and how
people uh
look at performance in roles this really
comes back to coaching again i mentioned
that earlier is
you know we've coaching came out of the
need to supervise
therapists and how they could be
guided in their path as they were
supposedly helping the people that were

needy

they needed guidance and that same idea

now of

coaching rather than teaching

or instructing and

um while they have their role as giving

the formal documents that you might need

and

providing the bigger narrative and

giving certain tools

you know you you within that you there's

this idea of

okay what types of developmental or

performance

interactions can be nurtured and

cultivated

and i think this actually really can tie

into the kind of pragmatist

social constructivist approaches that

coaching's trying to work with

as in but it gives it a bit more um

does actually give it a bit more meat

without becoming too complex which i

think

previously a lot of the active

influences may be a bit too abstract

but this is starting to show a bridge

there of why coaching

at different levels can and positive

psychology more broadly

can have a role

okay great thank you stephen dean yeah

stephen i appreciate exactly what you're

saying i

i think to get to a place where

interactionism

is no longer threatening

people have to want or appreciate that

you can be a subject matter expert and a
prediction matter expert
and i'm not sure i think there are maybe
are some
coaches who are both
and i but i sort of transcended that
and got into the wayfaring thing because
wayfaring imposes
a prediction aspect to this now i can't
predict for you
whether that running drum shot that you
make i can't be in your ear
i cannot coach you to take that shot in
that moment with
that much time left on the clock in fact
if i do that
i'm probably adding impediments and i'm
actually
being an optimal control list
micromanaging you
and you whether you make it or not it's
not about
me it's about whether the ball and the
and the and the basket arrived at some
sort of outcome that we both want
i think that i think that that's part of
how we get people to a place where
their dependency on their subject
subject matter expertise
doesn't overwhelm the fact that for
these performances
there is a predictive element and so we
have to
sort of give that equal billing i don't
think we have to over bill one or the
other
it's back to the between and that's a
prediction matter expert and

again i don't know if it's a role being
away
fairer because you're in the boat with
that other person right
it doesn't matter whether you're the one
saying okay well i'm
i'm getting a sense of the direction of
the current right now and the other
person's
looking up at the stars you're working
together
and so you everybody brings their
expertise
and i think that that only
that only materializes if we can if we
can include something with the subject
matter expertise
culturally yes absolutely there's huge
cultural piece to it
but i think i think you can be a coach
and not necessarily be a wayfarer
but i think if you're a wayfarer you
have to bring
a predictive element into it otherwise
you're not being a wayfarer
interesting about having an integrated
way to talk about
goal and process
and interaction with a niche interaction
with your co
wayfarers just like your example with a
movie
you know there's is the map app going to
tell you which
directions to take that's kind of on the
instructionist end
versus are you on the run so anywhere
you go

you're you're just not following
instructions you're just negotiating
sequential interactions and reducing
uncertainty in a way that
isn't just looking down at the
application and seeing which way to turn
when nice i'm glad you picked up on the
uh
the map going out the back end over the
trunk of the car
exactly oh you're reading it oh no
right um cool well
steven yes yeah just one question way
fairer is that like a particular sort of
approach is like an
orientation based or well yeah
yeah you're on your path and how are you
faring how are you feeling
okay right it's not just way finding
it's not just
the discovery piece but it's so how does
that make you feel being in that
uncertainty space
because again if we're going to close
loop discovery will eventually loop back
around to recapitulation
and then all right we're open systems
that are constantly closing these loops
so how does that make you feel right
okay
to connect that also to john boyk
and second order science as well as a
bucky fuller in
trim tab when we're
the rudder we need to act in an
anticipatory
fashion so that's kind of like first
order cybernetics the

rudder has to turn before you hit the
iceberg
you know you gotta find a path so
pathfinding and
that whole notion of anticipatory
control
comes to the rudder and the trim tab is
the little
rudder on the rudder so it's
anticipating
even the anticipatory nature
of planning because you have to turn
that trim tab
even before you turn the rudder and so
here we have instead of a sort of
mechanical model
with recursive levels of rudders we
actually have hierarchically nested
priors and so it's a little bit like
higher and higher level priors
that in one sense when you look at one
iceberg
and you try to tell a story by
backfilling what happens
they do act in increasingly anticipatory
ways
but then when it's seen as just part of
the wayfinding of the ship
they're all in a cyclic feedback with
each other and it's not like one is
really upstream of another
but there are certainly pieces doing
different things
cool stuff
any um final thoughts or questions i
don't see any other
questions from the live chat but
another um interesting and surprising

as usual does anyone have any
final comments
well great we'll um look forward to
some awesome upcoming streams let's just
look at those really fast
next week on the 21st
we're going to have our symposium with
carl fristen
so if you've been helping to co-organize
this event and develop the questions
with us
you'll be with us live otherwise we're
going to be recording this
not live streaming it and then providing
the recordings on our youtube
channel and then for the second two
weeks
of june we're going to be
doing a more technical or statistical
paper
empirical evaluation of active inference
in multi-armed bandits
then in early july we're going to have
a fields and living paper scale free
biology
integrating evolutionary and
developmental thinking
then uh don't have any papers for the
end of july or beginning of august but
we'll
figure out what we want to do there and
then we'll be
turning to some industrial engineering
and some other areas so
as always pretty fun journey
thanks uh dean blue steven
dave for participating live and everyone
for

watching live on replay we'll see you
on a future stream